TCS CORPORATE WEBSITE

Accordion Block — Developer Guide

Role / audience	Development — Developers, technical leads, block owners, AI coding agents
Version	1.0
Status	Draft for review
Owner	Technical Lead / Block Owner (proposed)
Last validated	2026-11-13
Documents	1
Classification	INTERNAL — classification to be confirmed

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Accordion Block — Developer Guide

Attribute	Value
Document ID	DEV-B-accordion
Audience	Developers, technical leads, block owners, AI coding agents
Purpose	How the Accordion block is built: files, model, structure, behaviour, variations, tests and limitations
Scope	blocks/accordion/ and its components accordion, accordion-item
Prerequisites	—
Owner	Technical Lead / Block Owner (proposed)
Approver	Product Owner (to be confirmed)
Version	1.0
Status	Draft for review
Last updated	13 November 2026
Last validated	2026-11-13 (model, files, imports from the code; reference pages from the live site)
Source files	blocks/accordion/README.md, blocks/accordion/_accordion.json, blocks/accordion/accordion.css, blocks/accordion/accordion.js
Related documents	Variations, Test coverage, Authoring guide
Related AEM paths	/content/tcs-corp/design-block-master/en/block-library/accordion, /content/tcs-corp/design-block-master/en/block-library/accordion/accordion

1. Summary

Attribute	Value
Block	Accordion
Folder	`blocks/accordion`
Type	Container
Category	interactive
Status	Active
Purpose	Expandable question / answer or detail panels (native details / summary)
Variations (summary)	Accordion Items; Additional tab options
Children	accordion-item (Accordion Item)
Code size	121 lines of JS and CSS

2. Overview

- **Business purpose:** present secondary or reference information (answers, process details, terms) compactly, so visitors scan the summaries and open only what they need while the full text stays in the page HTML.
- **User experience:** a stack of rows between thin dividers; each summary has a chevron on the right that turns when the item opens. Several items can be open at the same time; nothing is open when the page loads.
- **Templates / page types:** any page; the block is offered in every Section, in Columns Container columns and in Tabs Container tabs.

- **Data source:** authored content only (Summary and Text of each item). No sheet, query index, taxonomy or placeholder is read for rendering.
- **Related blocks:** Tabs Container (content compared or switched in place), Text (content that must always be visible), Fragment (to reuse one Accordion on several pages).
- **When to use:** FAQs, eligibility criteria, programme or process details, policy clarifications, long lists of capabilities with explanations.
- **When not to use:** key messages, prices, calls to action or legal notices every visitor must see (collapsed content is easily overlooked); a single item (use a Title and Text); items that need images, videos or nested blocks (the body is a rich text field).
- **Realistic use cases:** a careers FAQ ("How do I apply for a role at TCS?", "Can I apply for more than one position at the same time?"); graduate programme details on a careers page; mobility support explained on an employee page.

3. Source files

File	Link
`README.md`	GitHub
`_accordion.json`	GitHub
`accordion.css`	GitHub
`accordion.js`	GitHub

4. Components, models and filters

Component	Title	Resource type	Template name	Model	Children allowed
`accordion`	Accordion	`core/franklin/components/block/v1/block`	Accordion	`accordion`	accordion-item
`accordion-item`	Accordion Item	`core/franklin/components/block/v1/block/item`	Accordion Item	`accordion-item`	—

5. Fields

The Additional tab fields (anchor, hide, padding, margin, animation) are common to the blocks and described in the content modeling guide.

5.1 Accordion (`accordion`)

Tab	Label	Property	Type	Required in model	Default	Options	Visibility

5.2 Accordion Item (`accordion-item`)

Tab	Label	Property	Type	Required in model	Default	Options	Visibility
—	Summary	`summary`	text	No	`""`	—	—
—	Text	`text`	richtext	No	`""`	—	—

"Required in model" is the model's `required` flag; whether a field is required in practice is stated in the authoring guide where it is known.

6. Dependencies

Import
`../../scripts/scripts.js`
`../../decorators/editor-item-decorator.js`

7. For Developers

```
blocks/accordion/
├── README.md
├── _accordion.json
├── accordion.css
└── accordion.js
```

Decorator/Utility	Purpose	Execution or dependency point
scripts/scripts.js	Page decoration (moveInstrumentation and shared page helpers)	`accordion.js`: { moveInstrumentation }
decorators/editor-item-decorator.js	`buildEditorItems`, `editorItemHint` (in Universal Editor, an item the block doesn't render yet, e.g. a CTA just added, as a dashed placeholder the author can select; styles in `styles/lazy-styles.css`; the hint loaded o	`accordion.js`: { buildEditorItems, editorItemHint }

- **JavaScript (`accordion.js`):** `decorate(block)` turns each item row into `details.accordion-item` > `summary.accordion-item-label` + `div.accordion-item-body`, moving the cells' child nodes and the editor instrumentation (`moveInstrumentation` on the row and on the summary cell).
- **CSS (`accordion.css`):** dividers, chevron (`mask: url('../../icons/chevron-down.svg')`), typography tokens, hover / focus states, reduced-motion rules. All selectors are scoped to `.accordion`.
- **Input DOM (what AEM delivers):** the block isn't key-value, so it starts with an **empty row for its Additional dialog tab**, then one row per Accordion Item with two cells (Summary, Text). In Universal Editor item rows carry `data-aue-resource` / `data-aue-model="accordion-item"`.

```
<div class="accordion anchor-careers-faq">
  <div><div></div></div>
  <div><div>How do I apply for a role at TCS?</div><div><p>Search the open roles …</p></div></div>
</div>
```

- **Rendered DOM:**

```
<div class="accordion anchor-careers-faq block" id="anchor-careers-faq">
  <details class="accordion-item">
    <summary class="accordion-item-label">How do I apply for a role at TCS?</summary>
    <div class="accordion-item-body"><p>Search the open roles …</p></div>
  </details>
</div>
```

- **Decoration flow:** rows whose first cell is empty (no text and no `img`, `picture` or `a`) are "unnamed": the tab row and items without a Summary. In Universal Editor (rows with `data-aue-resource`) they are replaced by `buildEditorItems` placeholders with the `itemEditorHint` text (loaded only then, `editorItemHint`); otherwise they are removed. The remaining rows become `details`; a row without a second cell gets an empty body.
- **Events and async behavior:** no listeners of its own; toggling is native. `decorate` awaits `editorItemHint`, which only fetches placeholders in the editor.
- **Error handling:** none needed; if the script fails, the rows stay as plain text (aem.js logs the failure).
- **Exported functions:** the default `decorate` only.
- **Parent / child contracts:** filter `accordion` allows only `accordion-item`; `decorators/motion-decorator.js` animates `:scope > .accordion-item` (staggered), so keep that class on the item elements.

- **Data-layer hooks:** none in the block; `decorators/data-layer-decorator.js` captures the `toggle` of any `details` inside a block.

8. Extending the Block

1. **Add a field or option:** edit `blocks/accordion/_accordion.json` (keep the Additional tab last, at most three short tabs), then run `npm run build:json` and `npm run lint`.
2. **Map it to rendering:** a block option should be a `classes_*` field (delivered as a block class, never shifts rows). A new item field adds a cell to every item row: read cells by position from the start and keep the Summary as the first cell, because the decoration treats an empty first cell as "no item".
3. **Delivery rules:** a new dialog tab on the block model adds another empty row (already skipped as unnamed); a tab on the item model adds an empty cell to each item row (use `withoutTabCells` from `utils/cells.js`); field names ending in Text, Title, Type, Alt or MimeType collapse into their base field; don't put sample content in the definition template.
4. **Responsive and accessibility:** keep `details` / `summary`; any "open by default" or "one open at a time" option needs the `open` attribute managed in JS and must keep keyboard behaviour.
5. **Data layer:** opening / closing is already tracked; add `data-cmp-*` attributes only if a new control needs its own interaction name.
6. **Tests and library:** add unit tests for any pure logic under `test/`; add the variation to the block library page `block-library/accordion/accordion`; update this README (`npm run build:docs` regenerates the generated sections); run `npm run build:modules` after changing imports.
7. **Backward compatibility:** existing items have two cells (Summary, Text); keep reading them in that order.

9. Responsive Behavior

The block CSS has no media queries; the sizes come from tokens in `styles/styles.css` that change at 900px.

Viewport	Layout	Image behavior	Interaction behavior
Below 900px (mobile, tablet portrait)	Full-width rows; summary 1.125rem (`--heading-font-size-m`), body 1rem (`--body-font-size-m`); long summaries wrap	No images	Tap or keyboard toggles an item
900px and wider	Same rows; summary 1.5rem, body 1.125rem	No images	Hover and focus colour the summary brand blue

10. Accessibility

- **Semantic HTML:** each item is a native `details` element with a `summary`; screen readers announce it as expandable / collapsed without extra ARIA.
- **Headings:** summaries are not headings and don't appear in the heading outline; introduce the Accordion with a heading above it.
- **Keyboard and focus:** Tab reaches each summary; Enter or Space toggles it (native behaviour). Focus shows a 2px brand outline offset by 0.125rem; hover and focus turn the summary brand-coloured.
- **Accessible names:** the summary text. Items without a summary are removed on the page, so no nameless control is rendered (audit D-C-20, corrected).
- **Decorative elements:** the chevron is a CSS mask on a pseudo-element and is not announced.
- **Reduced motion:** the colour and chevron transitions are disabled with `prefers-reduced-motion: reduce`; the content animation doesn't play for reduced motion (AGENTS.md "Animation and skeletons").
- **Contrast:** colours come from tokens (`--color-text-default`, `--color-text-brand`, `--color-text-neutral-secondary`, `--color-border-default`); see Known Limitations for dark sections.

- **Error / loading / empty states:** none; the content is in the page HTML.
- **Reflow:** single column at every width; long summaries wrap.

Author responsibilities:

- Write summaries that make sense out of context; add a heading above the Accordion.
- Use descriptive link text in the answers; don't hide essential information in a closed item.

Developer responsibilities:

- Keep the native `details` / `summary` structure (it provides the semantics and keyboard support).
- Keep the focus style and the reduced-motion rules; use tokens for colours.

Key	Expected behavior
Tab / Shift+Tab	Moves to the next / previous summary (and to links inside open bodies)
Enter	Opens or closes the focused item
Space	Opens or closes the focused item

11. Data Layer and Analytics

- **Component:** registered by `decorators/data-layer-decorator.js` (`trackBlocks`) after the sections load: id = the block's id (Anchor ID), else `accordion-{n}` in page order; `@type` `tcs-corp/components/accordion`; `parentId` the page (or the containing block); `dc:title` only if the block contains a heading (normally none).
- **Variation:** none.
- **Item positions:** not recorded (the items have no `data-cmp-position`; the summary text identifies the item).
- **Impressions:** no impression event; only the component state above.
- **Interactions:** the `toggle` event of each `details` is captured centrally; summary clicks are excluded from `cmp:click`, so a toggle is counted once.

Trigger	Data-layer event	Required properties	Analytics behavior
Block decorated (page load)	component state	`component.{id}`: `@type`, `parentId`	Inventory of the blocks on the page
Item opened	`cmp:show`	`eventInfo.path` `component.{id}`, `dc:title` = summary text	Which questions visitors open
Item closed	`cmp:hide`	`eventInfo.path` `component.{id}`, `dc:title` = summary text	Closing interactions
Link in an answer clicked	`cmp:click`	`interaction` `link`, `dc:title`, `xdm:linkURL`, `xdm:linkType` (`other`, `exit` or `download`), `breakpoint`	Link tracking within the answers

No event is pushed when the block is inside an element with `data-analytics="off"`. The intended flow is Adobe Client Data Layer → Adobe Data Collection (Tags) → AEP Web SDK → Edge Network → Analytics / CJA, but Tags and the Web SDK aren't configured yet (blocked: measurement plan), so the events are only in `window.adobeDataLayer`. No consent mechanism is implemented.

12. Performance Considerations

- Loaded with its section (eager in the first section, else lazy), like any block; no dynamic imports.
- No network requests on the page; in Universal Editor the placeholders are fetched only when an empty item exists.
- No images; no listeners or observers of its own (the toggle is native; the data layer listens once on the document).
- DOM size grows with the number of items; closed bodies stay in the DOM.

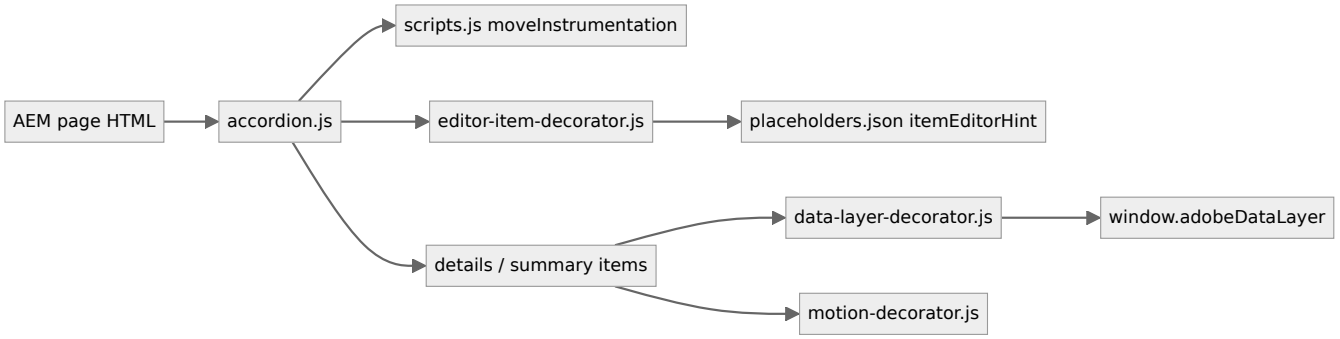
- Layout shift: none from the block; opening an item pushes the content below down by design.

13. Security and Privacy

- Content is authored HTML moved from the delivered cells (no `innerHTML` from strings).
- Links in answers are authored links; the block doesn't validate or rewrite them.
- No external content, no query strings read, no data stored; data-layer events contain only the summary text.

14. For Architects

- **Role:** a presentational disclosure block with authored content; no data source.
- **Content model:** a container block (`core/franklin/components/block/v1/block`) with item children (`.../block/item`, model `accordion-item`: Summary text, Text rich text); block model with the Additional tab only.
- **Rendering:** client-side decoration of server-delivered HTML; content is in the page HTML (indexable even when closed).
- **Dependencies:** `scripts/scripts.js` (`moveInstrumentation`), `decorators/editor-item-decorator.js`; centrally `decorators/data-layer-decorator.js`, `decorators/motion-decorator.js`, `decorators/block-options-decorator.js`.
- **Data layer / Web SDK / Target:** events in the Adobe Client Data Layer; Tags / Web SDK not configured; no Target.
- **Consent and identity:** none implemented; the events carry no personal data (summary text only).
- **Caching / performance:** delivered with the page through the Edge Delivery CDN; no requests of its own (outside the editor).
- **Security:** no external content; authored nodes are moved, not parsed from strings.
- **Scalability:** DOM size grows linearly with items; no limit in code.



15. Troubleshooting

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
Items show as plain text, no chevrons	Block script failed or wrong block	Developer tools: items should be `details.accordion-item`; console for `failed to load` messages; Network for `accordion.js`	Reload; escalate to development if the files fail	Items open and close
An item is missing on the page	Its Summary is empty	Check the item in the content tree; in the editor it is a dashed placeholder	Fill the Summary	The item renders
An item opens to nothing	Text empty	Select the item, check Text	Enter the answer or delete the item	The body shows

Symptom	Likely cause	Diagnostic steps	Resolution	Verification
Block missing on the live page	Hide block on, or page / fragment not published	Additional tab; publication status	Turn Hide block off; publish page and fragment	Block visible on aem.live
`#anchor-…` link does not reach the block	Anchor ID differs or was suffixed `-2`	Inspect the block's `id`	Correct or de-duplicate the Anchor ID	The link scrolls to the block
No `cmp:show` events	Inside `data-analytics="off"`, or Tags not reading them	Open an item, run `window.adobeDataLayer.slice(-3)` in the console	If present, configure Tags (not yet set up); else escalate	`cmp:show` appears

Evidence to collect (no secrets): page URL, browser and viewport, a screenshot, the block's rendered HTML from the developer tools, console errors, and the last data-layer entries.

16. Known limitations of the block

Limitation	Impact	Workaround	Planned correction
No "open by default", "one open at a time" or expand / collapse all option	All items start closed and open independently	Put essential content outside the Accordion	Not recorded
No FAQ structured data (JSON-LD)	No FAQ rich results	None	Not recorded
Summaries are not headings; no heading level option	Not in the heading outline	Add a Title above the block	Not recorded
Bodies are rich text only; no nested blocks or image styling	No media in items	Use other blocks outside the Accordion	Not recorded
The body text token (`--color-text-neutral-secondary`) and the divider token are not redefined by the dark-background theme contract in `styles/styles.css`	Body text contrast on Dark / Dark green sections may be insufficient	Check on `block-library/foundations/section-colours-with-blocks` before using it on dark sections	Not recorded

17. Change history of the block

Version	Change	Source	Date
Not recorded	README generated from verified implementation and approved handover documentation	Repository review	Not recorded
Handover 2.18	Additional tab Content animation (19 blocks, repeated items staggered), including the Accordion	docs/handover revision history	18 October 2026
Handover 2.27	Block ecosystem audit: the Accordion no longer renders an empty item from the dialog tab row	docs/handover revision history	19 October 2026
Handover 2.31	Re-audit fixes checked on the live pages, including the Accordion items	docs/handover revision history	20 October 2026

18. Reference pages

Kind	Page	Live URL
Block library	/design-block-master/en/block-library/accordion	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/accordion
Block library	/design-block-master/en/block-library/accordion/accordion	https://main--tcs-corp--sradha-tcs.aem.live/design-block-master/en/block-library/accordion/accordion

19. Coverage of the required topics

Topic	Where
Purpose	Overview
Source files	generated: Files
Model, definition, filters, children	generated: Components
Dependencies	generated: Dependencies
Authored input / rendered output / DOM transformation	For Developers
Configuration fields, defaults, required fields	generated: Fields
Variations	ACCORDION-VARIATIONS.md
Responsive behaviour	Responsive Behavior
Accessibility and keyboard	Accessibility
Analytics / data layer	Data Layer and Analytics
Performance	Performance Considerations
Security	Security and Privacy
Extending	Extending the Block
Known limitations	Known Limitations
Troubleshooting	Troubleshooting
Tests	ACCORDION-TEST-COVERAGE.md
SEO behaviour	Not recorded
Caching considerations	Not recorded (Edge Delivery default caching; no block-specific rule)
Ownership	Technical Lead / Block Owner (proposed)

20. Screenshots

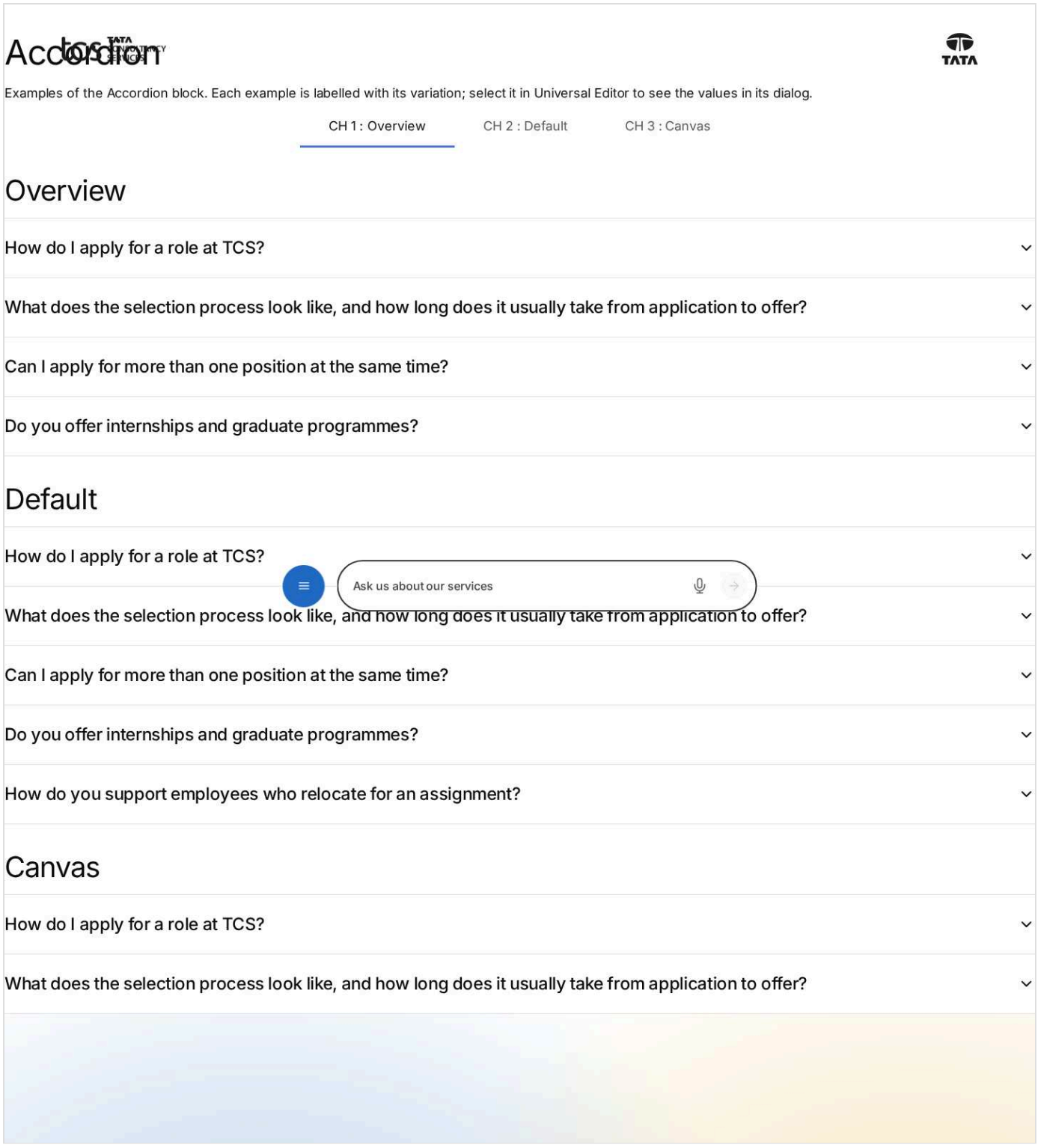


Figure: Accordion block library page — desktop 1440 × 900 (/design-block-master/en/block-library/accordion/accordion, captured 2026-11-08).

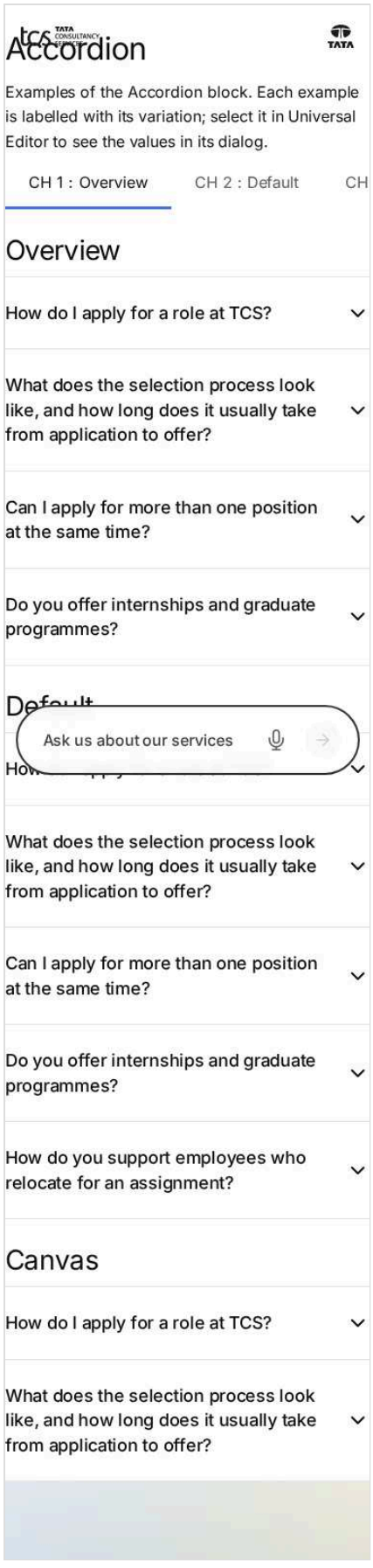


Figure: Accordion block library page — mobile 375 × 812 (/design-block-master/en/block-library/accordion/accordion, captured 2026-11-08).

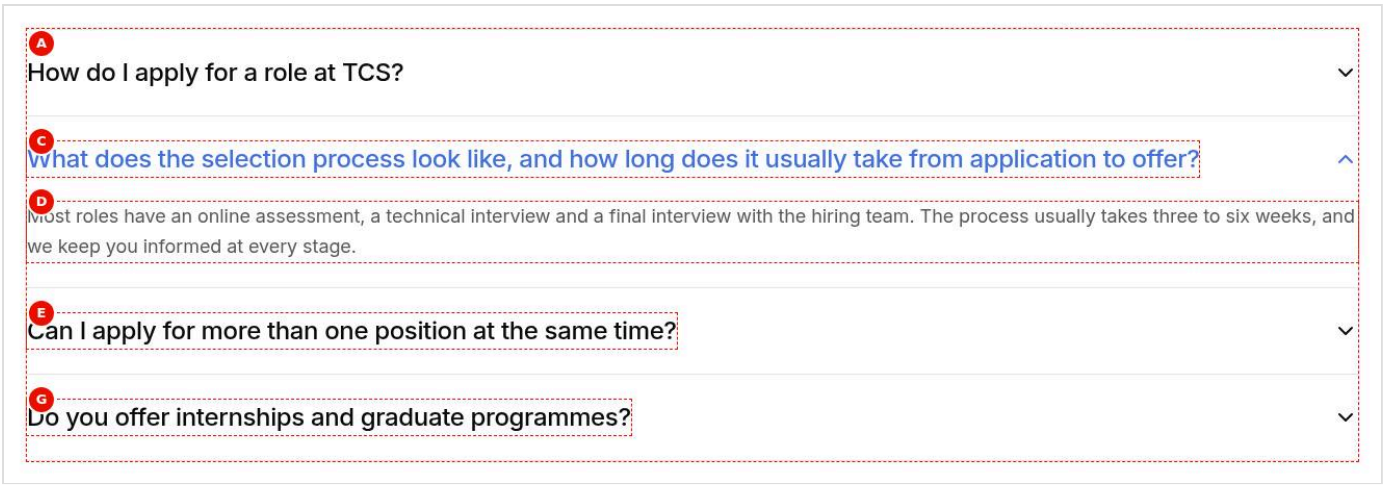


Figure: Desktop rendering — The Accordion at 1440 px: four items 'How do I apply for a role at TCS?', the selection-process question, 'Can I apply for more than one position at the same time?' and 'Do you offer internships and graduate programmes?' between 1px dividers, chevrons on the right; the second item open with its chevron turned up and its answer about the online assessment and interviews below. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/accordion-desktop-view.jpg) (handover screenshot; rendering at the time it was taken).

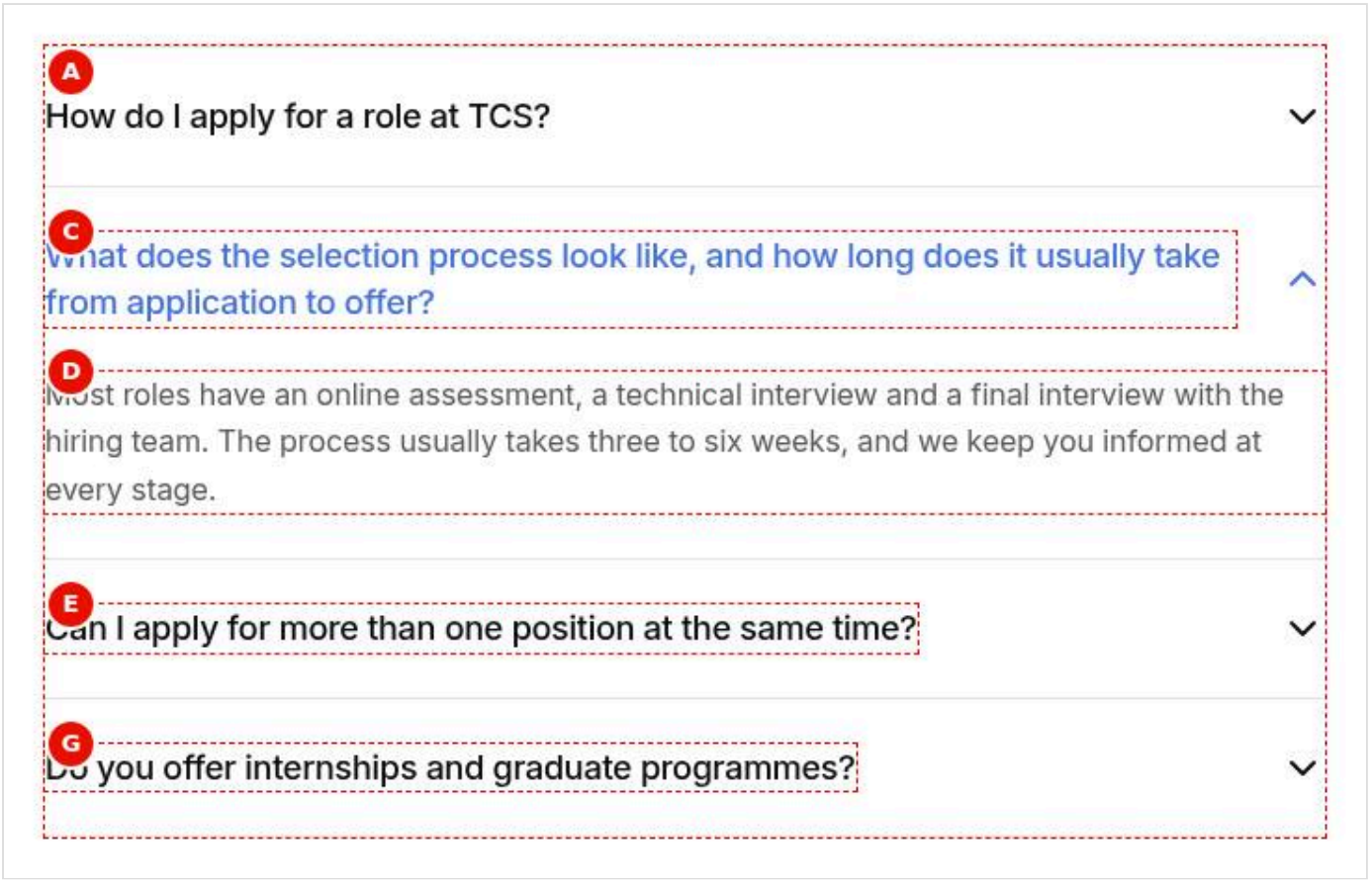


Figure: Tablet rendering — The Accordion at 768 px with the same layout as on desktop but smaller type (summary 1.125rem, body 1rem): the four items 'How do I apply for a role at TCS?', the selection-process question, 'Can I apply for more than one position at the same time?' and 'Do you offer internships and graduate programmes?'; the second, two-line question open with its answer below. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/accordion-tablet-view.jpg) (handover screenshot; rendering at the time it was taken).

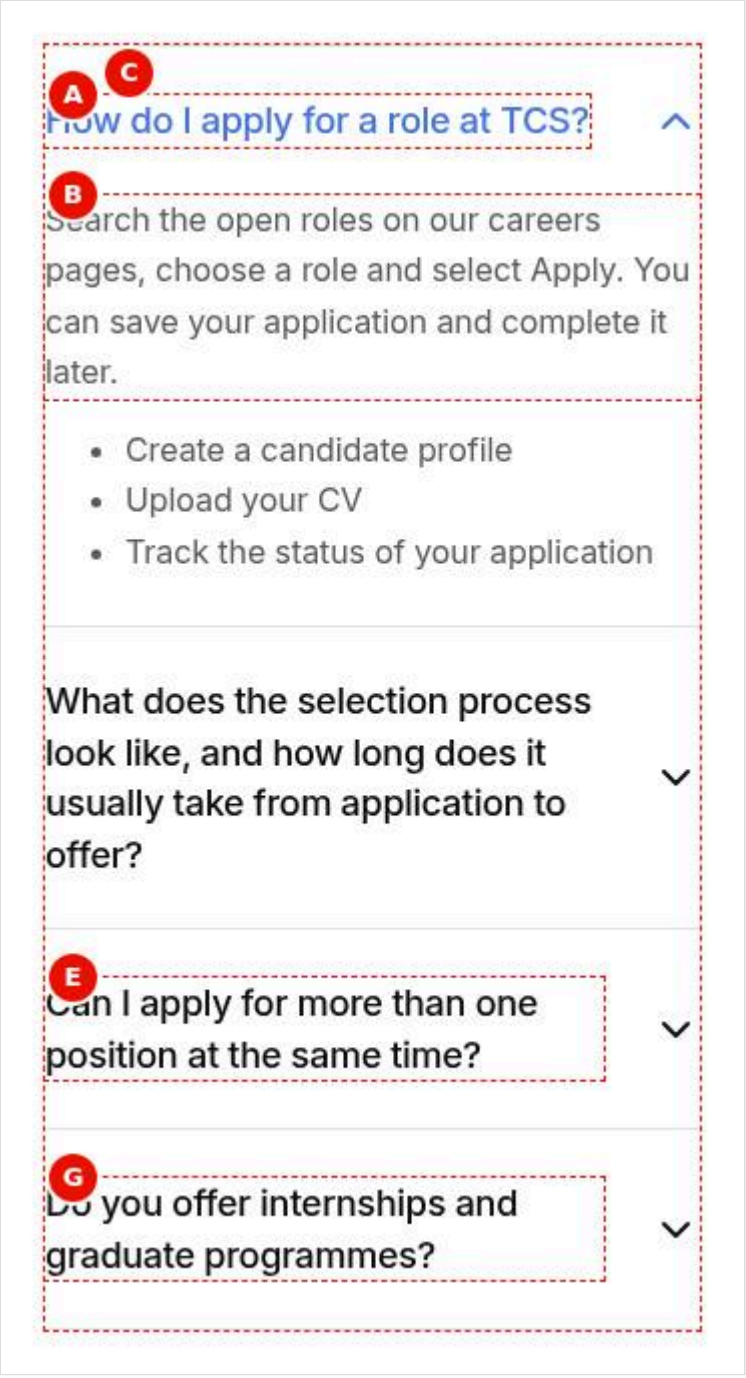


Figure: Mobile rendering — The Accordion at 375 px: the first item 'How do I apply for a role at TCS?' open with its 1rem paragraph and three bullets (create a candidate profile, upload your CV, track the status of your application); the three closed summaries below wrap at 1.125rem with chevrons on the right. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/accordion-mobile-view.jpg) (handover screenshot; rendering at the time it was taken).

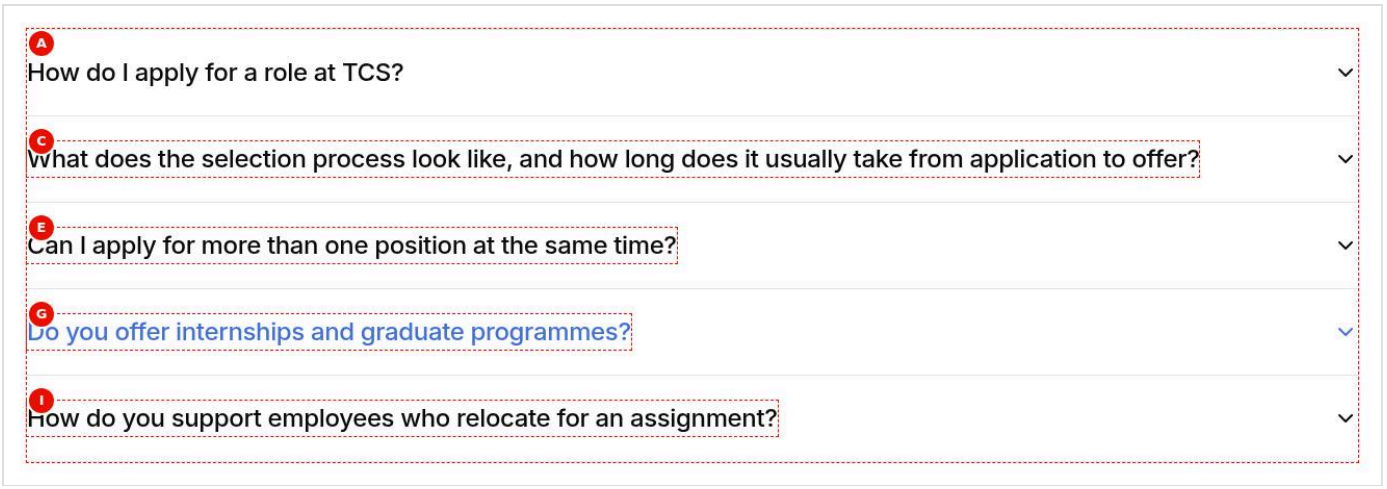


Figure: Default Accordion variation — The Accordion at 1440 px with five closed items between 1px dividers, each with a downward chevron: the four careers questions and 'How do you support employees who relocate for an assignment?'; the third summary, under the pointer, in brand blue. (Sample content on the preview; callout letters as in the Field Entry Table; file screenshots/accordion-default.jpg) (handover screenshot; rendering at the time it was taken).

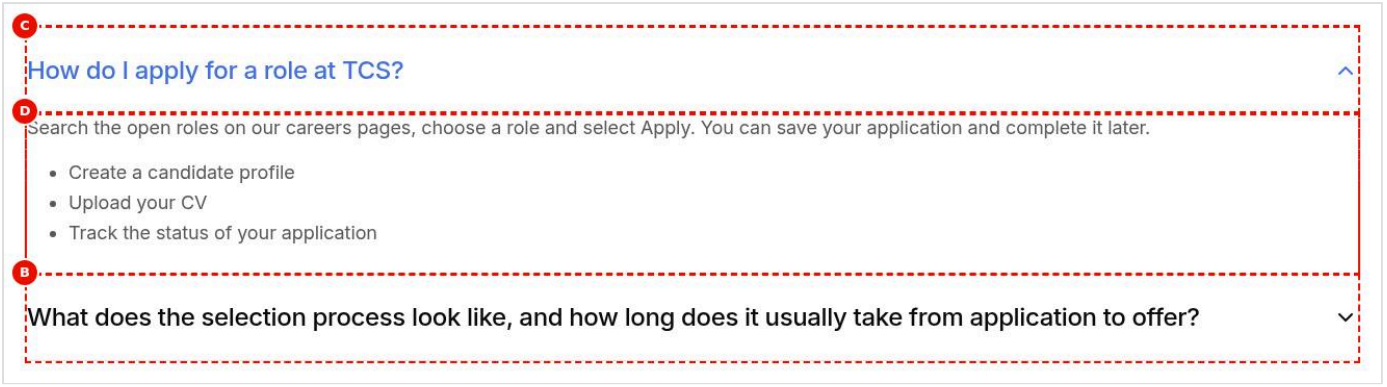


Figure: Universal Editor canvas — The Accordion at 1440 px in Universal Editor Edit mode (annotated preview, not a Universal Editor screenshot) with two items: A on the block, B on the second item, C on the first summary 'How do I apply for a role at TCS?' and D on its open text with the bulleted list. (Annotated preview in editor mode with sample content; letters as in the annotations list, not a Universal Editor screenshot; file screenshots/accordion-authoring-view.jpg) (handover screenshot; rendering at the time it was taken).

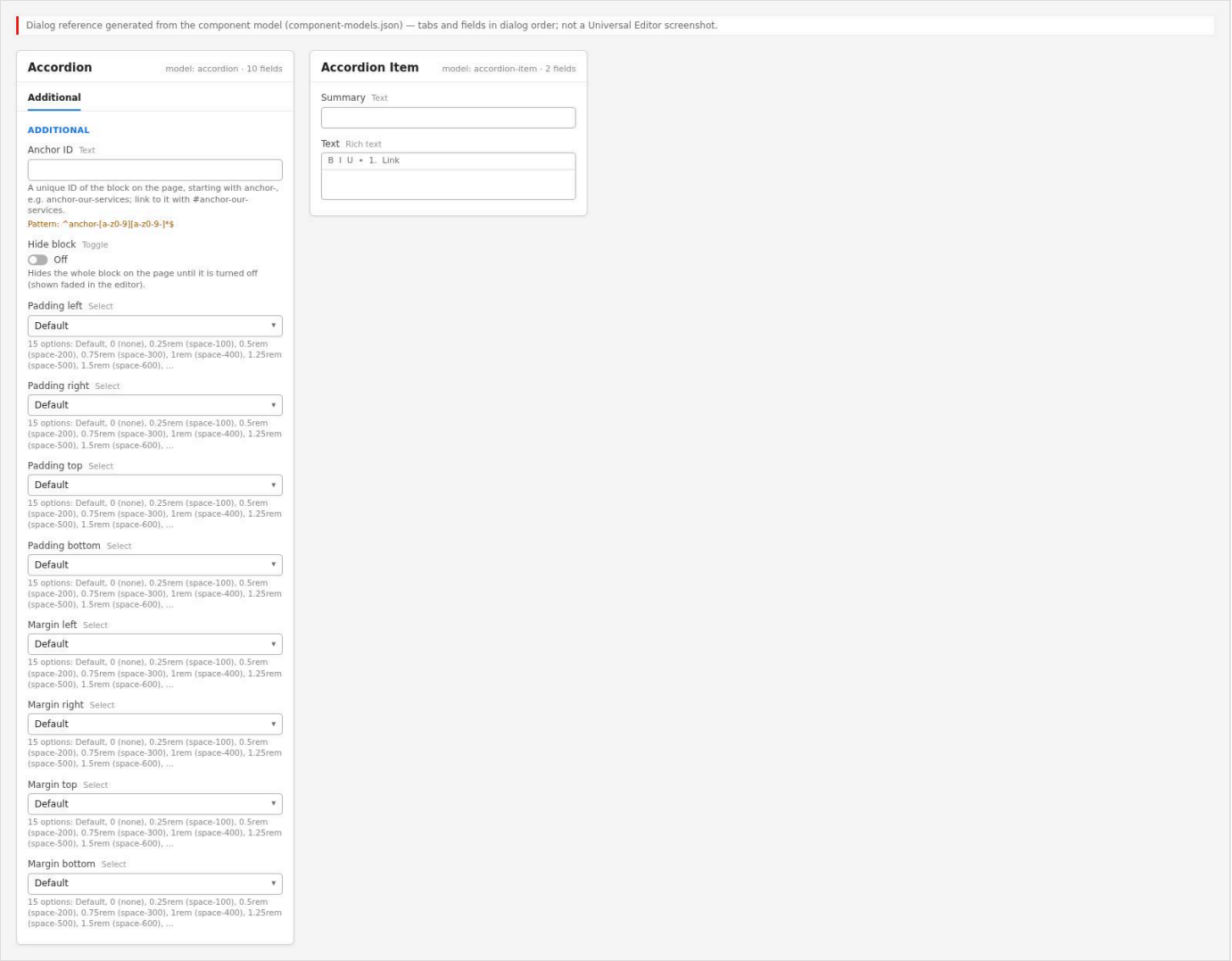


Figure: Properties panel — The Universal Editor properties panel for the Accordion block (Additional tab with Anchor ID, Hide block and the eight spacing selects) and, side by side, for an Accordion Item (Summary and Text). (Dialog reference generated from the component model, not a Universal Editor screenshot; file screenshots/accordion-properties-panel.jpg) (handover screenshot; rendering at the time it was taken).

21. Known limitations

- Universal Editor dialogs were not captured: no author access in this program (see the access blockers).

22. Open items

- None.

23. Change history

Version	Date	Author	Change
1.0	13 November 2026	Delivery team (documentation program)	First version of the role-specific documentation (master documentation prompt), generated from the repository and the published reference pages